1 A student uses a Hall probe to investigate the magnetic flux density due to a U-shaped electromagnet, as shown in Fig. 1.1.

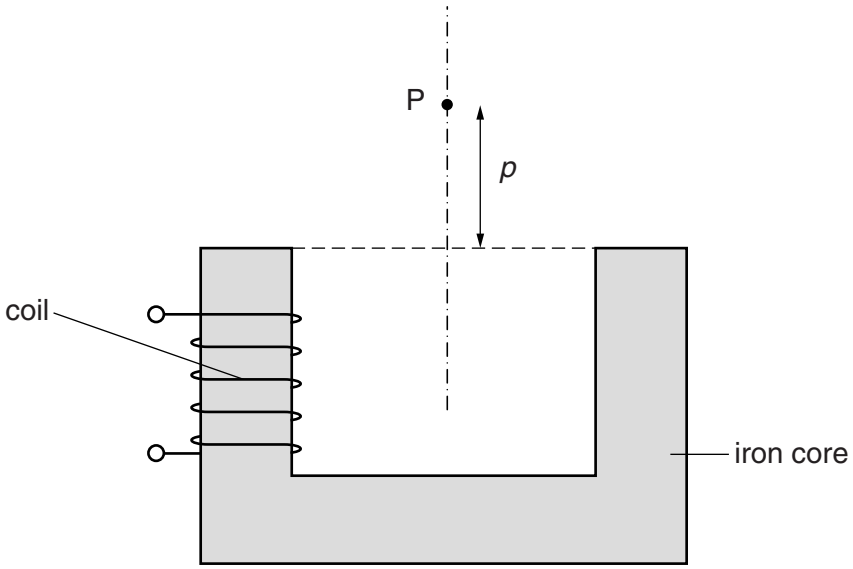

Fig. 1.1

Point P is equidistant from the poles of the electromagnet and distance p is the vertical distance between P and the top of the electromagnet. The magnetic flux density is B at point P.

It is suggested that the relationship between B and p is

$$B = kNIe^{-\alpha p}$$

where N is the number of turns on the coil, I is the current in the coil and α and k are constants.

Design a laboratory experiment using a Hall probe to test the relationship between B and p. Explain how your results could be used to determine values for α and k. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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